Linear Statistical Models - W9&10 GA

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1. The dataset '*InsectSpray*' has data on the count of insects in areas treated with one of 6 different types of sprays.
   Note: The dataset is already in the proper format for the one-way analysis of variance – a vector with the data (count), and one with a factor describing the level (spray).
   Based on the given information, answer the following questions:

   (a) Import dataset in *R*, using the following command: [1 Mark]
   require(datasets); data(InsectSprays)

   (b) Using *R*, plot a side-by-side box plot to see if the treatment means are equal. [3 Marks]

   (c) Perform a one-way ANOVA to check if the treatment means are equal. Do they agree? [5 Marks]

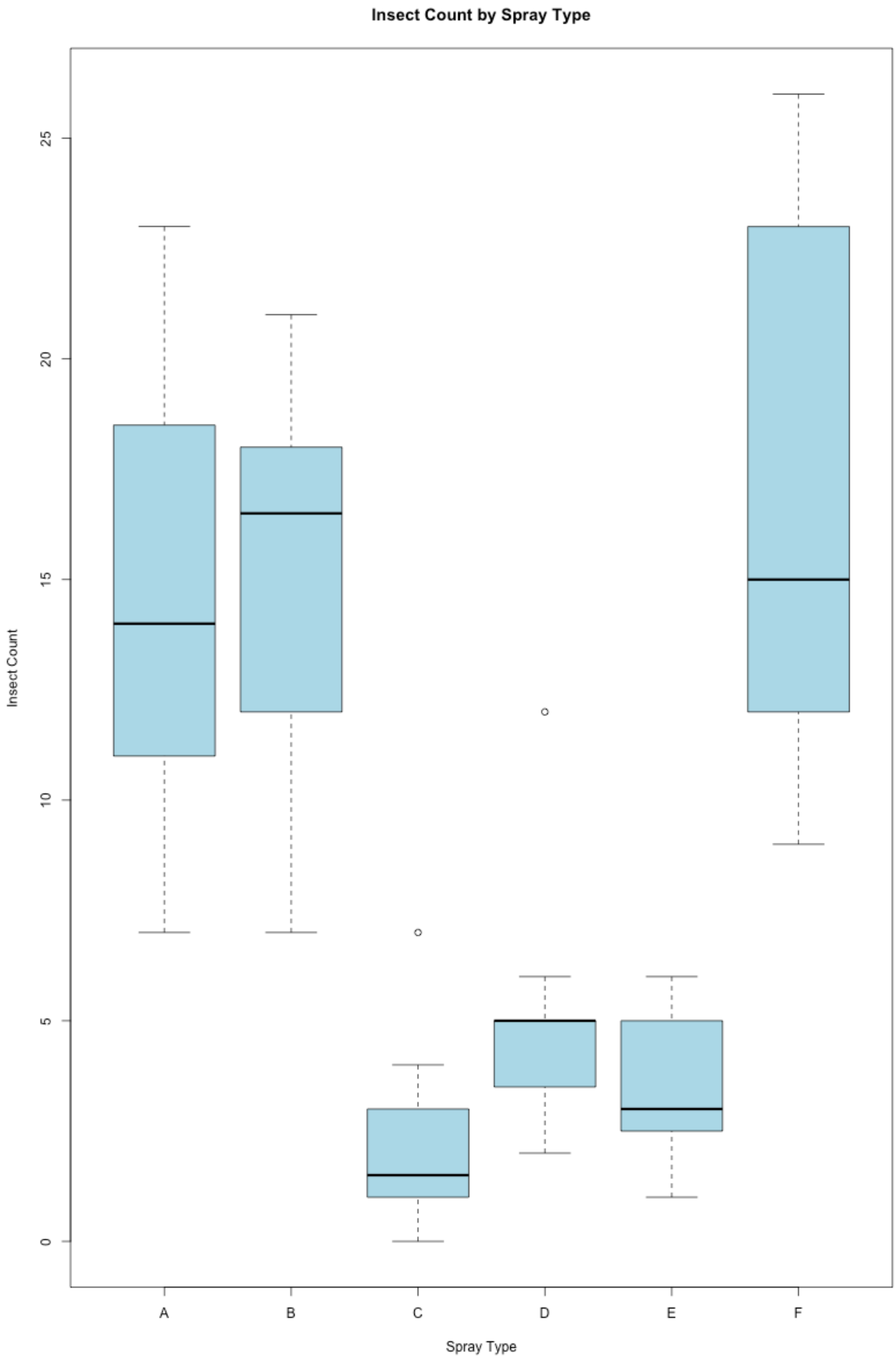
a)

```
require(datasets)
data(InsectSprays)
head(InsectSprays)
```

```
r$> require(datasets)
    data(InsectSprays)
    head(InsectSprays)
  count spray
1    10     A
2     7     A
3    20     A
4    14     A
5    14     A
6    12     A
```

b)

```r
boxplot(count ~ spray,
   data = InsectSprays,
   main = "Insect Count by Spray Type",
   xlab = "Spray Type",
   ylab = "Insect Count",
   col = "lightblue"
)
```



c)

```r
anova_model <- aov(count ~ spray, data = InsectSprays)
summary(anova_model)

# Since p-value(<2e-16 ~approx. 0) < alpha we reject the null hypothesis
# and conclude that the treatment means are not equal.
```

```
r$> anova_model <- aov(count ~ spray, data = InsectSprays)
    summary(anova_model)
            Df Sum Sq Mean Sq F value Pr(>F)
spray        5   2669   533.8    34.7 <2e-16 ***
Residuals   66   1015    15.4
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

2. A manufacturing company has purchased three new machines of different types, say, m_1, m_2 and m_3. Owner of the company wants to analyse the effectiveness of the machines for which he wants to observe the five outputs from each machine.

(a) Simulate 50 positive values from discrete uniform distribution for machine m_1. Repeat it for m_2 and m_3 by varying parameters of the distribution. [2 Marks]

```r
set.seed(123) # Set seed for reproducibility

n <- 50

m1 <- sample(10:20, n, replace = TRUE) # Machine 1 values
m2 <- sample(15:25, n, replace = TRUE) # Machine 2 values
m3 <- sample(20:30, n, replace = TRUE) # Machine 3 values
```

```
r$> print(m1)
    print(m2)
    print(m3)
 [1] 12 12 19 11 15 20 14 13 15 18 19 20 14 12 20 18 18 18 12 17 19 16 19 18 12 13 10 20
[29] 16 14 19 16 18 18 19 16 20 14 16 14 20 15 18 11 14 17 11 10 18 20
 [1] 23 20 19 23 24 18 20 25 22 20 20 21 15 20 16 15 16 18 19 20 17 23 18 20 23 23 21 17
[29] 22 23 17 21 17 21 20 24 19 19 22 17 24 16 24 16 24 20 18 15 20 17
 [1] 27 22 27 20 26 30 26 26 29 25 26 30 29 24 25 27 24 26 30 23 22 28 26 25 29 28 26 21
[29] 22 27 23 26 23 20 27 23 28 27 25 30 23 27 22 23 23 25 20 29 30 23

r$> 
```

(b) Consider the following models:

- $y_{ij} = m_i + \epsilon_{ij}$
- $y_{ij} = 3 + 4m_i + \epsilon_{ij}$
- $y_{ij} = -1 + 2m_i + \epsilon_{ij}$

Note: Simulate error terms, ϵ_{ij} from the $N(0, 1)$ distribution and use the simulated values of part (a).

Store the values in the data frame with two columns, i.e. Machine type and Effectiveness score (y_{ij}'s). [3 Marks]

```r
epsilon <- rnorm(n * 3, mean = 0, sd = 1)

# Create a combined factor vector for Machine Type
machine_type <- factor(rep(c("m1", "m2", "m3"), each = n))

# Combine the 'm' values into a single vector for calculation
m_values <- c(m1, m2, m3)

# --- Model 1: y = m + epsilon ---
y1 <- m_values + epsilon
data_model1 <- data.frame(Machine = machine_type, Effectiveness = y1)

# --- Model 2: y = 3 + 4m + epsilon ---
y2 <- 3 + (4 * m_values) + epsilon
data_model2 <- data.frame(Machine = machine_type, Effectiveness = y2)

# --- Model 3: y = -1 + 2m + epsilon ---
y3 <- -1 + (2 * m_values) + epsilon
data_model3 <- data.frame(Machine = machine_type, Effectiveness = y3)
```

```r
r$> epsilon <- rnorm(n * 3, mean = 0, sd = 1)

    # Create a combined factor vector for Machine Type
    machine_type <- factor(rep(c("m1", "m2", "m3"), each = n))

    # Combine the 'm' values into a single vector for calculation
    m_values <- c(m1, m2, m3)

    # --- Model 1: y = m + epsilon ---
    y1 <- m_values + epsilon
    data_model1 <- data.frame(Machine = machine_type, Effectiveness = y1)

    # --- Model 2: y = 3 + 4m + epsilon ---
    y2 <- 3 + (4 * m_values) + epsilon
    data_model2 <- data.frame(Machine = machine_type, Effectiveness = y2)

    # --- Model 3: y = -1 + 2m + epsilon ---
    y3 <- -1 + (2 * m_values) + epsilon
    data_model3 <- data.frame(Machine = machine_type, Effectiveness = y3)
```

(c) Define null and alternative hypotheses to perform one-way ANOVA. [3 Marks]

```r
null_hypo <- "The mean effectiveness scores for all three machines are equal."

alternate_hypo <- "At least one machine's mean effectiveness
score is different from the others."
```

$$H_0 : \mu_{m1} = \mu_{m2} = \mu_{m3}$$

(d) For each of the model, write an R - code (from scratch) to perform one-way ANOVA and comment on the obtained result about acceptance/ rejection of hypothesis. [10 Marks]

```r
perform_anova_scratch <- function(df) {
  # 1. Calculate Grand Mean
  grand_mean <- mean(df$Effectiveness)

  # 2. Calculate Group Means
  group_means <- aggregate(Effectiveness ~ Machine, df, mean)
```

```r
# 3. Calculate SS_Between (Treatment Sum of Squares)
SS_Between <- 0
for (lvl in group_means$Machine) {
    ni <- sum(df$Machine == lvl)
    mi <- group_means$Effectiveness[group_means$Machine == lvl]
    SS_Between <- SS_Between + (ni * (mi - grand_mean)^2)
}

# 4. Calculate SS_Total (Total Sum of Squares)
SS_Total <- sum((df$Effectiveness - grand_mean)^2)

# 5. Calculate SS_Error (Residual Sum of Squares)
SS_Error <- SS_Total - SS_Between

# 6. Degrees of Freedom
k <- length(unique(df$Machine)) # Number of groups (3)
N <- nrow(df) # Total observations (150)
df_between <- k - 1
df_error <- N - k

# 7. Mean Squares
MS_Between <- SS_Between / df_between
MS_Error <- SS_Error / df_error

# 8. F-Statistic
F_stat <- MS_Between / MS_Error

# 9. P-value
p_value <- 1 - pf(F_stat, df_between, df_error)

# Output Results
cat("--- ANOVA From Scratch ---\n")
cat("F-Statistic:", F_stat, "\n")
cat("P-value:", p_value, "\n")

if (p_value < 0.05) {
    cat("Result: Reject H0 (Means are different)\n\n")
```

```r
    } else {
        cat("Result: Accept H0 (Means are equal)\n\n")
    }
}

# Run for all 3 models
print("Model 1 Analysis:")
perform_anova_scratch(data_model1)

print("Model 2 Analysis:")
perform_anova_scratch(data_model2)

print("Model 3 Analysis:")
perform_anova_scratch(data_model3)
```

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```
    y3 <- -1 + (2 * m_values) + epsilon
    data_model3 <- data.frame(Machine = machine_type, Effectiveness = y3)

    perform_anova_scratch(data_model1)

    print("Model 2 Analysis:")
    perform_anova_scratch(data_model2)

    print("Model 3 Analysis:")
    perform_anova_scratch(data_model3)
[1] "Model 1 Analysis:"
--- ANOVA From Scratch ---
F-Statistic: 108.7607
P-value: 0
Result: Reject H0 (Means are different)

[1] "Model 2 Analysis:"
--- ANOVA From Scratch ---
F-Statistic: 128.8834
P-value: 0
Result: Reject H0 (Means are different)

[1] "Model 3 Analysis:"
--- ANOVA From Scratch ---
F-Statistic: 123.3838
P-value: 0
Result: Reject H0 (Means are different)
```

Comment: **We reject the Null hypothesis for all 3 models, as the p-value is less than alpha (0.05)**. Hence, we conclude that **"Means are different."**

(e) For each of the above-mentioned models, perform one-way ANOVA by using inbuilt function in R. Comment on the obtained result and compare with the outputs obtained in part (e). [8 Marks]

```
check_builtin <- function(df, model_name) {
   cat("--- Built-in ANOVA for", model_name, "---\n")
   res <- aov(Effectiveness ~ Machine, data = df)
   print(summary(res))
   cat("\n-----------------------------------\n")
}

check_builtin(data_model1, "Model 1")
check_builtin(data_model2, "Model 2")
check_builtin(data_model3, "Model 3")
```

```
    check_builtin(data_model1, "Model 1")
    check_builtin(data_model2, "Model 2")
    check_builtin(data_model3, "Model 3")
--- Built-in ANOVA for Model 1 ---
             Df Sum Sq Mean Sq F value Pr(>F)
Machine       2   2252  1126.1   108.8 <2e-16 ***
Residuals   147   1522    10.4
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

-----------------------------------
--- Built-in ANOVA for Model 2 ---
             Df Sum Sq Mean Sq F value Pr(>F)
Machine       2  36386   18193   128.9 <2e-16 ***
Residuals   147  20750     141
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

-----------------------------------
--- Built-in ANOVA for Model 3 ---
             Df Sum Sq Mean Sq F value Pr(>F)
Machine       2   9067    4534   123.4 <2e-16 ***
Residuals   147   5401      37
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

-----------------------------------
```

Comments:

The F-stat and results are the same in both parts (d) and (e)

Model	F-Statistic (Built-in)	F-Statistic (Scratch)	Result Status
Model 1	108.8	108.7607	**MATCH**
Model 2	128.9	128.8834	**MATCH**
Model 3	123.4	123.3838	**MATCH**

The small differences (e.g., 108.8 vs 108.7607) are due to standard rounding in the R summary display.

Here also, we can conclude that **there is statistically significant evidence to conclude that the effectiveness scores (y_{ij}) of the three machines are different.** This is consistent with our simulation setup, where you assigned different value ranges for each machine (e.g., 10-20 vs. 20-30).